

12.120

EE25BTECH11002 - Achat Parth Kalpesh

Question:

Let $\mathbf{A}$ be a 4×4 matrix with eigen values $-5, -2, 1, 4$. Which of the following is the eigenvalue of

$$\begin{pmatrix} \mathbf{A} & \mathbf{I} \\ \mathbf{I} & \mathbf{A} \end{pmatrix}, \quad (0.1)$$

where $\mathbf{I}$ is the 4×4 identity matrix ?

1) -5

2) -7

3) 2

4) 1

Solution:

Let $\mathbf{v}$ be any eigenvector of $\mathbf{A}$ with eigenvalue λ , i.e.

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v} \quad (4.1)$$

and

$$\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{I} \\ \mathbf{I} & \mathbf{A} \end{pmatrix} \quad (4.2)$$

Let

$$\mathbf{x} = \begin{pmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{pmatrix} \quad (4.3)$$

be an eigenvector of $\mathbf{M}$ corresponding to eigenvalue μ where $\mathbf{x}_1$ and $\mathbf{x}_2$ are 4×1 column vectors, i.e.

$$\mathbf{M}\mathbf{x} = \mu\mathbf{x} \quad (4.4)$$

Substituting,

$$\begin{pmatrix} \mathbf{A} & \mathbf{I} \\ \mathbf{I} & \mathbf{A} \end{pmatrix} \begin{pmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{pmatrix} = \begin{pmatrix} \mu\mathbf{x}_1 \\ \mu\mathbf{x}_2 \end{pmatrix} \quad (4.5)$$

This gives two matrix equations:

$$\mathbf{A}\mathbf{x}_1 + \mathbf{x}_2 = \mu\mathbf{x}_1 \quad (4.6)$$

$$\mathbf{x}_1 + \mathbf{A}\mathbf{x}_2 = \mu\mathbf{x}_2 \quad (4.7)$$

From (4.6),

$$(\mathbf{A} - \mu\mathbf{I})\mathbf{x}_1 + \mathbf{x}_2 = 0 \quad (4.8)$$

$$\mathbf{x}_2 = -(\mathbf{A} - \mu\mathbf{I})\mathbf{x}_1. \quad (4.9)$$

Substitute this in (4.7):

$$\mathbf{x}_1 + \mathbf{A} [- (\mathbf{A} - \mu \mathbf{I}) \mathbf{x}_1] = \mu [- (\mathbf{A} - \mu \mathbf{I}) \mathbf{x}_1] \quad (4.10)$$

$$\mathbf{x}_1 - \mathbf{A} (\mathbf{A} - \mu \mathbf{I}) \mathbf{x}_1 = -\mu (\mathbf{A} - \mu \mathbf{I}) \mathbf{x}_1 \quad (4.11)$$

$$[\mathbf{I} - (\mathbf{A} - \mu \mathbf{I})^2] \mathbf{x}_1 = 0 \quad (4.12)$$

For a non-trivial solution,

$$\det [\mathbf{I} - (\mathbf{A} - \mu \mathbf{I})^2] = 0 \quad (4.13)$$

As

$$\mathbf{A} \mathbf{v} = \lambda \mathbf{v} \quad (4.14)$$

$$[\mathbf{I} - (\mathbf{A} - \mu \mathbf{I})^2] \mathbf{v} = [\mathbf{v} - (\mathbf{A} - \mu \mathbf{I})^2 \mathbf{v}] \quad (4.15)$$

Let

$$(\mathbf{A} - \mu \mathbf{I})^2 \mathbf{v} = (\mathbf{A} - \mu \mathbf{I}) (\mathbf{A} - \mu \mathbf{I}) \mathbf{v} \quad (4.16)$$

$$(\mathbf{A} - \mu \mathbf{I}) \mathbf{v} = \mathbf{A} \mathbf{v} - \mu \mathbf{v} \quad (4.17)$$

$$= (\lambda - \mu) \mathbf{v} \quad (4.18)$$

$$(\mathbf{A} - \mu \mathbf{I}) (\lambda - \mu) \mathbf{v} = (\lambda - \mu)^2 \mathbf{v} \quad (4.19)$$

$$[\mathbf{I} - (\mathbf{A} - \mu \mathbf{I})^2] \mathbf{v} = [1 - (\lambda - \mu)^2] \mathbf{v} \quad (4.20)$$

For a non-zero $\mathbf{v}$, this holds only if

$$1 - (\lambda - \mu)^2 = 0 \quad (4.21)$$

$$\mu = \lambda \pm 1 \quad (4.22)$$

Given eigenvalues of $\mathbf{A}$ are

$$\lambda \in \{-5, -2, 1, 4\} \quad (4.23)$$

Therefore, the eigenvalues of $\mathbf{M}$ are

$$\lambda + 1 \in \{-4, -1, 2, 5\} \quad (4.24)$$

$$\lambda - 1 \in \{-6, -3, 0, 3\} \quad (4.25)$$

Eigenvalues of $\mathbf{M}$ are $-6, -4, -3, -1, 0, 2, 3, 5$